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Experiment-37

Question

Write a **menu-driven C program** to perform the following operations using **separate functions**:

1. Count spaces, digits, and special characters in a string.
2. Count frequency of a given character in a string.
3. Toggle the case of each character in a string.
4. Exit.

Algorithm

1. Start.
2. Declare the required variables and function prototypes.
3. Display the menu:
 1. Count spaces, digits, and special characters.
 2. Count frequency of a given character.
 3. Toggle the case of each character.
 4. Exit.
4. Read the user's choice.
5. Use a `switch` statement to perform the selected operation.
6. If the choice is **1**:
 - Read a string.
 - Initialize `spaces = 0`, `digits = 0`, and `special = 0`.
 - Traverse the string character by character.
 - If the character is a space, increment `spaces`.
 - If the character is a digit from 0 to 9, increment `digits`.
 - If the character is neither an alphabet, digit, space, nor newline, increment `special`.
 - Display the number of spaces, digits, and special characters.
7. If the choice is **2**:
 - Read a string.
 - Read the character whose frequency is to be found.
 - Initialize `count = 0`.
 - Traverse the string character by character.
 - If the current character is equal to the given character, increment `count`.
 - Display the frequency of the given character.
8. If the choice is **3**:

- Read a string.
 - Traverse the string character by character.
 - If a character is uppercase, convert it to lowercase.
 - If a character is lowercase, convert it to uppercase.
 - Leave digits, spaces, and special characters unchanged.
 - Display the modified string.
9. If the choice is **4**, display an exit message and terminate the program.
 10. If the choice is other than 1, 2, 3, or 4, display "**Invalid choice**".
 11. Repeat the menu until the user selects option 4.
 12. Stop.

Test Case Table

Test Case	Choice	Input String	Additional Input	Expected Output
TC1	1	Hello 123@#	—	Spaces = 1, Digits = 3, Special characters = 2
TC2	1	C Program 2026!	—	Spaces = 2, Digits = 4, Special characters = 1
TC3	2	banana	a	Frequency of a = 3
TC4	2	Hello World	l	Frequency of l = 3
TC5	2	programming	z	Frequency of z = 0
TC6	3	Hello WORLD	—	hELLO world
TC7	3	AbC123@xYz	—	aBc123@XyZ
TC8	4	—	—	Exiting program
TC9	5	—	—	Invalid choice

C Program

```
#include <stdio.h>
```

```
void countCharacters();
```

```
void frequency();
```

```
void toggleCase();
```

```
int main()
```

```
{
```

```
    int choice;
```

```
    do
```

```
    {
```

```
        printf("\n--- MENU ---\n");
```

```
        printf("1. Count spaces, digits and special characters\n");
```

```
        printf("2. Count frequency of a character\n");
```

```
        printf("3. Toggle case of each character\n");
```

```
        printf("4. Exit\n");
```

```

printf("Enter your choice: ");
scanf("%d", &choice);
getchar();

switch(choice)
{
    case 1:
        countCharacters();
        break;

    case 2:
        frequency();
        break;

    case 3:
        toggleCase();
        break;

    case 4:
        printf("Exiting program.\n");
        break;

    default:
        printf("Invalid choice.\n");
}

} while(choice != 4);

return 0;
}

void countCharacters()
{
    char str[100];
    int i, spaces = 0, digits = 0, special = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ' ')
            spaces++;

        else if(str[i] >= '0' && str[i] <= '9')
            digits++;

        else if(!((str[i] >= 'A' && str[i] <= 'Z') ||

```

```

        (str[i] >= 'a' && str[i] <= 'z') ||
        str[i] == '\n'))
    special++;
}

printf("Spaces = %d\n", spaces);
printf("Digits = %d\n", digits);
printf("Special characters = %d\n", special);
}

```

```

void frequency()
{
    char str[100], ch;
    int i, count = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    printf("Enter character to search: ");
    scanf("%c", &ch);
    getchar();

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ch)
            count++;
    }

    printf("Frequency of '%c' = %d\n", ch, count);
}

```

```

void toggleCase()
{
    char str[100];
    int i;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] >= 'A' && str[i] <= 'Z')
            str[i] = str[i] + 32;

        else if(str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - 32;
    }
}

```

```
    printf("String after toggling case: %s", str);  
}
```

Compilation And Execution

```
(kali㉿kali)-[~]  
$ vi hdoc37.c  
  
(kali㉿kali)-[~]  
$ gcc hdoc37.c -o hdoc37  
  
(kali㉿kali)-[~]  
$ ./hdoc37  
  
— MENU —  
1. Count spaces, digits and special characters  
2. Count frequency of a character  
3. Toggle case of each character  
4. Exit  
Enter your choice: 1  
Enter a string: Hello 123@#  
Spaces = 1  
Digits = 3  
Special characters = 2  
  
— MENU —  
1. Count spaces, digits and special characters  
2. Count frequency of a character  
3. Toggle case of each character  
4. Exit  
Enter your choice: 2  
Enter a string: banana  
Enter character to search: a  
Frequency of 'a' = 3
```

```
— MENU —  
1. Count spaces, digits and special characters  
2. Count frequency of a character  
3. Toggle case of each character  
4. Exit  
Enter your choice: 3  
Enter a string: abC123@xYz  
String after toggling case: ABc123@XyZ  
  
— MENU —  
1. Count spaces, digits and special characters  
2. Count frequency of a character  
3. Toggle case of each character  
4. Exit  
Enter your choice: 4  
Exiting program.
```